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RESEARCH PAPER

Mid-IR laser measurement of acetaldehyde in the headspace gas of cell culture media samples from growth of breast cancer cells in hydrogel scaffolds

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Abstract
A mid-IR laser absorption spectrometer configured with an interband cascade laser (ICL) was used to measure acetaldehyde concentrations in the headspace gas of cell culture media samples obtained from the culturing of breast cancer cells in a flow perfusion bioreactor. Measurements were performed by bubbling lab air through a media sample and passing the exhaust gas through a long optical path gas cell where rotational-vibrational modes of acetaldehyde were excited by the ICL. Acetaldehyde concentrations were determined from peak-to-peak voltages for the two strongest acetaldehyde absorption features within a spectral region spanning 1770.20 cm^{-1} to 1770.35 cm^{-1}. Three different media samples obtained from cells cultured with the same nominal conditions had headspace acetaldehyde concentrations of 412 ppb, 513 ppb, and 390 ppb, while two media samples with imposed hypoxia or cobalt chloride stabilization of hypoxia inducible factor 1-alpha (HIF-1α) had higher acetaldehyde concentrations, 815 ppb and 536 ppb, respectively. These results establish experimental proof-of-concept for the ability of mid-IR laser absorption spectroscopy to measure acetaldehyde in cell culture media headspace, allowing observation of HIF-1α driven shifts in cancer cell metabolism.

Keywords Cancer cell metabolism · Breast cancer ell cultures · Flow perfusion bioreactor · Acetaldehyde · Hypoxia inducible factor 1-alpha · Laser absorption spectroscopy · Mid-infrared · Interband cascade laser

Introduction

Cancer cells are known to have an altered metabolism that results in the production of specific small molecule metabolites that enter the gas phase. The sensing of these gas-phase metabolites can provide a new way to detect cancer, a capability that has already been demonstrated by multiple studies showing accurate cancer detection by canine olfaction [1–3]. Recently, advances in the development of mid-infrared (mid-IR) lasers for chemical sensing applications have created an opportunity to significantly improve the state of the art in gas-phase metabolite sensing. This paper describes the development and use of a trace gas sensor based on an interband cascade laser (ICL) with tunable emission in the 5.65 micron spectral region, which is within the R-branch of the carbonyl stretch mode of vibration for acetaldehyde. This laser is used to measure acetaldehyde by exciting quantized vibrational modes, which are also coupled to closely spaced quantized rotational modes, to give absorption spectra that are uniquely specific to the acetaldehyde molecule. Ultimately, this sensor technology, which does not require reagents or other consumables, will be easy to use and can be manufactured at scale for low-cost production. It offers promise for use in cancer research, cancer therapy monitoring, and early cancer detection.

Cancer cell metabolism is altered from healthy cell metabolism by overexpression of hypoxia-inducible factor 1-alpha (HIF-1α) [4], a DNA-binding protein with transcriptional activities that include gene encodings for glucose transporters, which increase glucose uptake, and activation of pyruvate dehydrogenase kinase-1 (PDK-1), which blocks pyruvate from entering the mitochondria for aerobic respiration [5]. Pyruvate concentrations in cancer cell cytosol are

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therefore higher than they are in healthy cells, an effect first observed by Otto Warburg as an increase in lactate concentration due to the enzyme lactate dehydrogenase (LDH) [6], which converts pyruvate to lactate. Higher pyruvate concentrations in cancer cells are also expected to result in higher acetaldehyde concentrations because pyruvate decomposes exothermally to acetaldehyde and carbon dioxide with an energy release of 19.9 to 21.6 kJ/mol (0.21 to 0.22 eV/molecule) depending on ionic strength, as shown by the energy of formation data in Table 1 [7]. Acetaldehyde has a high vapor pressure at 37 C, above 1.8 atmospheres with its 20.2 C boiling point. It also has a high activity coefficient in aqueous solutions [8], so it will have relatively high equilibrium vapor phase concentrations. These thermochemical properties make acetaldehyde an attractive molecule for measurement in the gas phase. A high acetaldehyde concentration in the vapor phase can indicate elevated pyruvate concentrations in cell cytosol and possible HIF-1α overexpression. Measurement of acetaldehyde concentrations in the headspace of cultured lung cancer cells by selected ion flow tube mass spectrometry (SIFT-MS) showed concentrations ranging between 82 ppb and 1.470 ppm depending on cell line and number of cultured cells [9, 10]. Additional SIFT-MS measurements of lung cancer cells cultured on hydrogel scaffolds showed gas-phase acetaldehyde concentrations as high as 1.973 ppm for 1 million cells [11]. Measurements of headspace gas from cultured HL60 human leukemia cancer cells using gas chromatography/mass spectrometry (GC/MS) have also been performed, and acetaldehyde concentrations were shown to increase from 490 to 698 ppb between day 1 and day 2 of a cell culture [12].

Laser absorption spectroscopic measurement of gas-phase acetaldehyde concentrations by tuning a mid-IR laser within the carbonyl stretch absorption band was first described by Kamat et al. [13]. In this prior work, a cryogenically cooled IV-VI semiconductor diode laser with tunable single-mode emission near 1727.1 cm^{-1}, within the P-branch of the carbonyl stretch band, was used with a 100-m optical path gas cell to measure acetaldehyde concentrations in gas-phase samples. This laser spectrometer demonstrated an acetaldehyde minimum detection limit in the range of 40 ppb. It was used to measure acetaldehyde concentrations in exhaled human breath following alcohol consumption. Exhaled acetaldehyde concentrations as high as 4 ppm were measured, an indication of ethanol metabolism since its metabolic pathway involves acetaldehyde as an intermediate byproduct. In this paper, we describe a mid-IR laser spectrometer equipped with a thermoelectrically cooled ICL with tunable single-mode emission between 1769.4 cm^{-1} and 1770.8 cm^{-1}, which is within the R-branch of the carbonyl stretch band. We also describe its use in measuring acetaldehyde concentrations in the headspace gas of cell culture media samples obtained from breast cancer cell cultures performed with a flow perfusion bioreactor. This paper provides a detailed description of this new ICL spectrometer, which has a minimum acetaldehyde detection limit of 37 ppb, thus closely matching the performance of the prior spectrometer. The demonstration of a mid-IR laser spectrometer with a thermoelectrically cooled ICL shows that it will be possible to develop compact and lightweight instrumentation for trace gas sensing applications. Such sensors will facilitate integration with bioreactors for on-line headspace monitoring, a possible future extension of the work described here, as well as facilitate cancer detection research involving analysis of exhaled breath, for example.

Materials and methods

Mid-IR laser spectrometer and headspace gas measurement

The state of the art in mid-IR laser technology has advanced significantly with the development of ICLs. Threshold currents for mid-IR ICLs, which are fabricated from semiconductor quantum well materials that have a type-II band alignment [14, 15], are more than an order of magnitude lower than those for quantum cascade lasers (QCLs). This is because the upper laser level in an interband laser is not depopulated by phonon scattering as easily as it is in QCLs. Recently, mid-IR ICLs have been used to sense methane and ethane near 3000 cm^{-1} [16], carbon dioxide near 2400

Table 1 Energies of formation, ΔG_f, for pyruvate ($C_3H_4O_3$), acetaldehyde (CH_3CHO), and CO_2 at three different ionic strengths, as expected in biological systems during adenosine triphosphate (ATP)–driven hydrolysis. The Gibbs free energies of formation, $\Delta G = \Delta H - T\Delta S$, for pyruvate decarboxylation, are calculated for a 37 °C temperature using data from ref. [7]

	Energy of formation ΔG_f, at 37 °C kJ/mol		
Molecule Ionic strength (mol/l):	0.00	0.10	0.25
Pyruvate ($C_3H_4O_3$)	− 342.6	− 341.3	− 340.9
Acetaldehyde (CH_3CHO)	30.2	32.8	33.6
Carbon dioxide (CO_2)	− 394.4	− 394.4	− 394.4
ΔG for pyruvate decarboxylation	− 21.6	− 20.3	− 19.9

cm^{-1} [17], nitrous oxide and carbon monoxide near 2200 cm^{-1}, and nitric oxide near 1930 cm^{-1} [18]. Here, we use a distributed feedback (DFB) ICL (Nanoplus GmbH) with tunable emission near 1770.25 cm^{-1} to sense acetaldehyde by exciting rotational-vibrational molecular motions within the R-branch of the carbonyl stretch band. Figure 1 shows emission spectra from this ICL, measured at a 1 cm^{-1} spectral resolution using a Fourier transform infrared (FTIR) spectrometer. With a constant 75 mA injection current, the emission frequencies blue shift from 1769.4 cm^{-1} to 1770.8 cm^{-1} as the heat sink temperature is cooled from − 5 to − 14 C. The emission from this DFB laser is predominantly

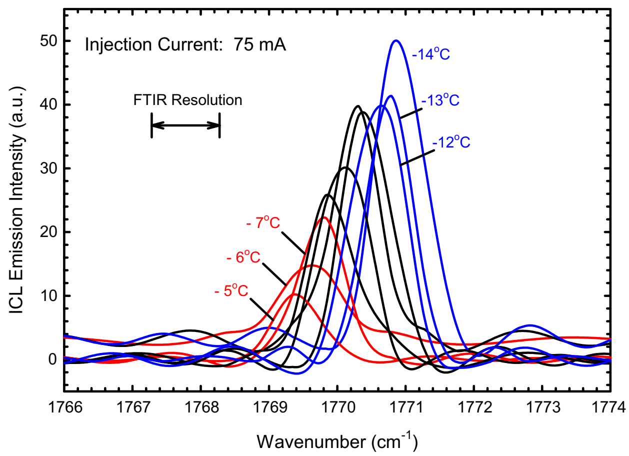


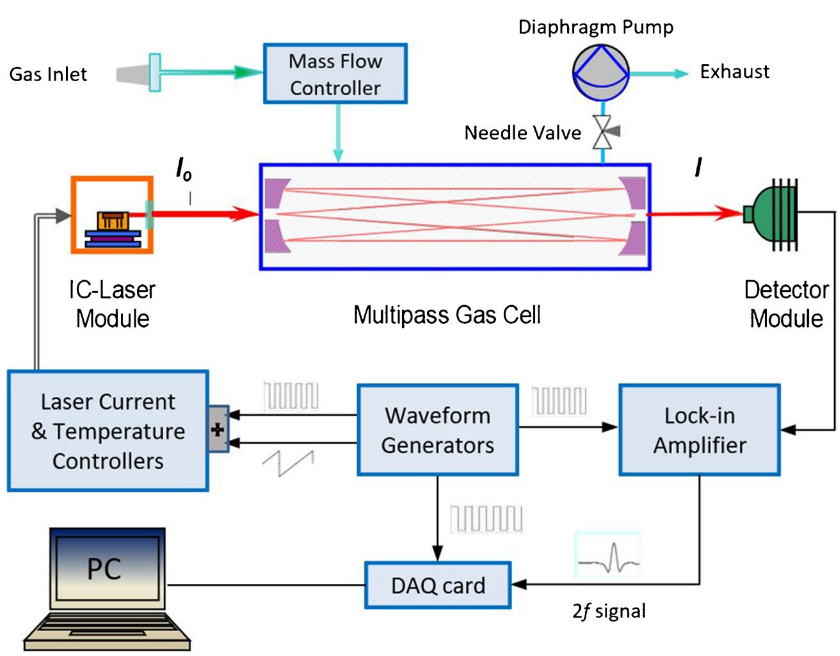
Fig. 1 ICL emission from − 5 to − 14 C in 1 C increments for an injection current of 75 mA as measured by an FTIR spectrometer with 1 cm^{-1} resolution. A dip in the emission intensity trend at − 12 C and − 13 C is due to a strong water absorption line at 1770.85 cm^{-1}

single-mode with less than 10% of the power in the side modes, which appear near 1768 cm^{-1} and 1772 cm^{-1}. This laser was packaged with a thermoelectric cooler (TEC) in a TO-66 can and assembled into a module with a short focal length lens for beam collimation.

Figure 2 is a block diagram showing the main components of the laser absorption spectrometer that was fabricated with this ICL. Electronics for driving the ICL consisted of an ILX Lightwave current source with an integrated TEC controller (model #LDC-3900) and two waveform generators (SRS model #DS345), one for tuning the ICL at 100 Hz with a sawtooth ramp (25 mv peak-to-peak) and one for high frequency modulation (40 kHz). The nominal ICL injection current was 78 mA with a − 13 C heat sink temperature. A HgCdTe photovoltaic detector with integrated TEC temperature control (Vigo model #PDI-2 TE-6) was used to measure ICL light intensity after passing through a 100-me optical path Herriott gas cell (Aerodyne). A series of planar mirrors directed the ICL beam through the gas cell and to the detector. A lock-in amplifier (SRS model #SR810 DSP), synced with the high-frequency ICL modulation, was used to sample the detector signal at 80 kHz. A desktop computer equipped with 100 kHz sampling rate data acquisition hardware (National Instruments, model #6052E) and software (LabView) was used to acquire, process, and store photodetector voltages during the ICL tuning ramp as well as peak-to-peak voltages for selected absorption features as a function of time. Measured absorption spectra consist of second harmonic detector signals acquired in 1000 channels over the 10 ms laser current ramp time.

Figure 3 shows absorption spectra for lab air, formic acid headspace, and a 1-in-thick germanium etalon with a free

Fig. 2 ICL-based gas sensor with a λ = 5.65 µm ICL. Major components include the ICL and HgCdTe detector modules (both with integrated thermoelectric coolers), a multipass gas cell, electronics, and a flow controller and diaphragm pump for gas sampling



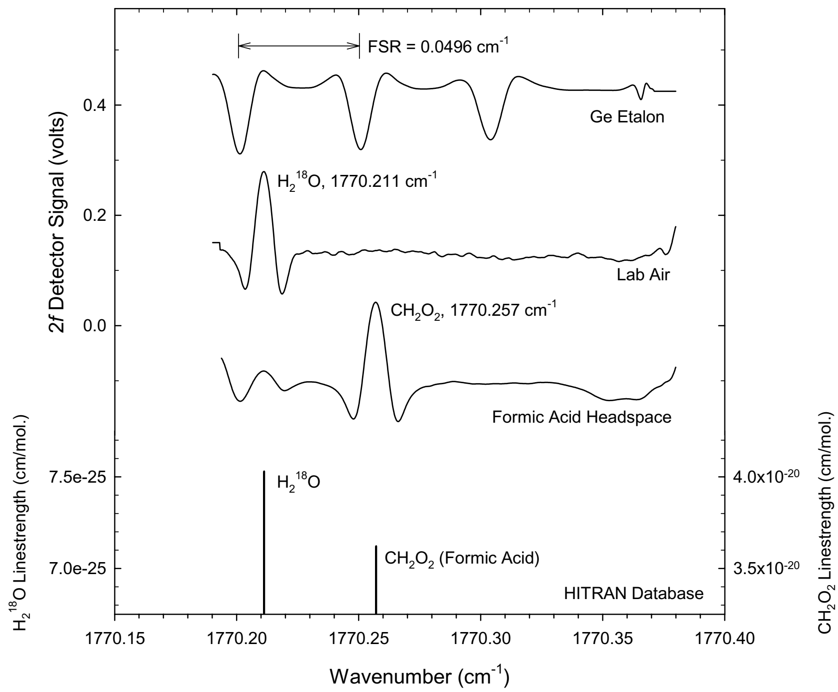


Fig. 3 Laser absorption spectra obtained with an ICL heatsink temperature of − 13 C and an injection current of 77 mA. Spectra for a germanium etalon with a free spectral range of 0.0496 cm^{-1}, lab air, and formic acid headspace are shown. Rotational-vibrational line positions from the Hitran database for an isotope of water, $H_2{}^{18}O$, and formic acid are also shown

spectral range of 0.0496 cm^{-1}. The lab air spectrum shows only one significant peak at 1770.211 cm^{-1}, which is due to absorption by a specific isotope of water, $H_2{}^{18}O$, according to the Hitran database [19]. The horizontal wavenumber axis in Fig. 3 was generated using this water absorption as a reference and the Ge etalon peaks for axis scaling. The spectrum for formic acid headspace shows a strong absorption peak at 1770.257 cm^{-1}, agreeing with the Hitran database value for formic acid absorption. These data confirm an ICL tuning range between 1770.19 and 1770.35 cm^{-1}. A 0.16 cm^{-1} tuning range divided into 1000 channels gives this spectrometer a spectral resolution in the range of 1.6×10^{-4} cm^{-1}.

Figure 4 shows absorption spectra for three different gas samples, 21 ppm acetaldehyde in a balance of nitrogen (Specgas, Inc.), lab air, and formic acid headspace, plotted as a function of channel number. The acetaldehyde spectrum exhibits three prominent peaks at 1770.305 cm^{-1}, 1770.283 cm^{-1}, and 1770.266 cm^{-1} according to the spectral calibration data in Fig. 3. Separated by about 0.020 cm^{-1} (600 MHz) from each other, these peak absorbances are consistent with the absorption fine structure created by the excitation of torsional modes of rotation within the acetaldehyde molecule [20]. The peak-to-peak voltages for the two strongest peaks are determined by subtracting the minimum second harmonic detector signal between the two peaks, near channel number 425, from the signals at the two peaks near channel numbers 350 and 475. The spectrum for the 21 ppm acetaldehyde gas sample shows peak-to-peak voltages of 282 mV and 211 mV for the strongest and second strongest peaks giving calibration factors of 74 ppb/mV and 100 ppb/mV, respectively. Figure 5 shows the Allan-Werle variance plot for measurement of the strongest acetaldehyde peak at 1770.305 cm^{-1}. This plot shows that instrument noise, both electrical and optical, can be as low as about 0.5 mV for an electronics integration time of about 30 s. This corresponds to a minimum acetaldehyde detection limit of 37 ppb.

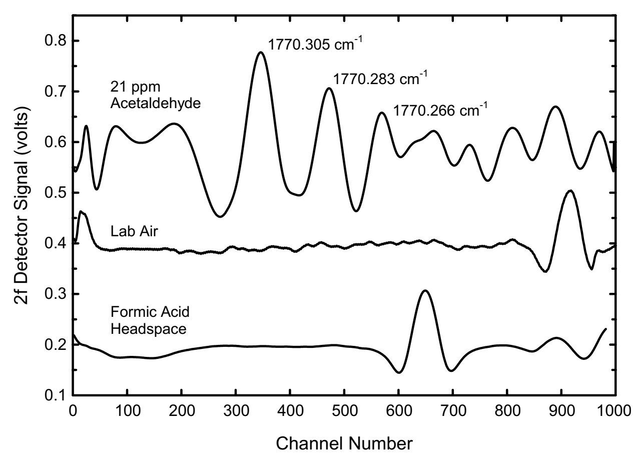


Fig. 4 Second harmonic absorption spectra for laboratory air, formic acid headspace, and 21 ppm acetaldehyde in N_2 plotted as a function of channel number

Mid-IR laser measurement of cell culture media headspace gas was performed by bubbling laboratory air through about 40 ml of culture media contained in a 50-ml flask. Exhaust gas from the flask flowed into a second 250-ml

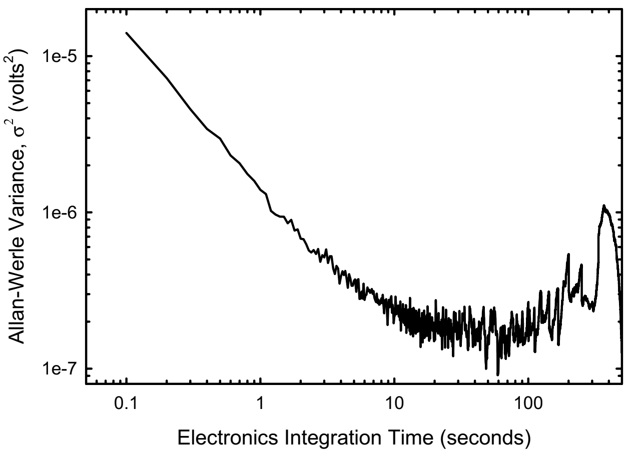


Fig. 5 Allan-Werle plot for measurement of acetaldehyde at 1770.305 cm^{-1}

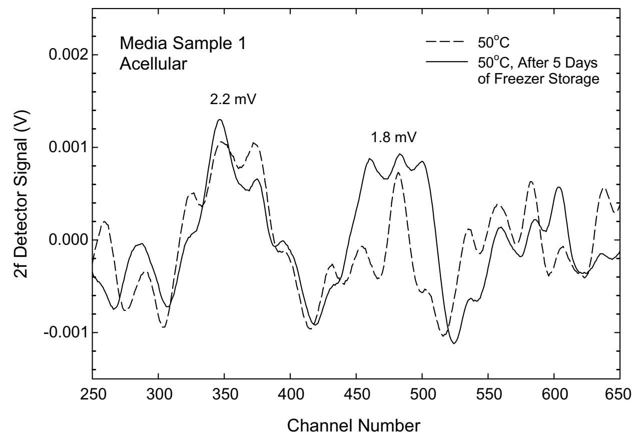


Fig. 6 Mid-IR laser absorption spectra on two different days for an acellular (unused) media sample at 50 C

flask to collect condensate, at times abundant when bubbles formed with some media samples. This flowing lab air, saturated with media sample headspace gas, was then passed through a 1-m coiled section of ¼″ outer diameter Teflon tubing immersed in ice water. It then entered the Herriott gas cell through a mass flow controller set at 100 ml/min. A diaphragm pump (Vacuubrand, model #MD8) induced gas flow from the exhaust of the gas cell, and a needle valve between the pump and the gas cell was set to maintain a pressure of 20 Torr in the gas cell. The media sample temperature, measured with a mercury thermometer immersed in the media, was stabilized in a heated water bath. About 2 min of continuous gas flow was required before peak-to-peak voltages for acetaldehyde reached steady values, which corresponded to the time needed to establish steady state headspace gas flow through the tubing and condensate traps. Measurement of 21 ppm acetaldehyde reference gas flowing through this system showed no significant difference in peak voltages compared to when the 21 ppm reference gas was introduced directly into the gas cell, showing that acetaldehyde was not being trapped in the ice water-immersed Teflon tubing.

Real-time spectra and real-time concentration trends for selected peaks in the spectra were observable on the computer screen, and spectra were saved at various times during a sample measurement. A typical media sample measurement involved saving a background spectrum for chilled lab air only immediately prior to initiating lab air flow through the media sample. Two sample spectra were then saved about 1 min apart from each other during steady-state flow conditions. Each saved spectrum consisted of the coadded sum of 3000 spectra acquired over 30 s of electronics integration time. In addition, trend charts for the peak-to-peak voltages of the two strongest acetaldehyde peaks and the water peak, near channel 900 in Fig. 4, were saved following the measurements. In this case, a LabView algorithm was used to find the minimum and maximum values within a spectral window around each peak, and the differences were recorded every 5 s, with each data point representing a running average of 3000 co-added spectra.

Figure 6 shows background-subtracted mid-IR laser absorption spectra for the headspace gas of an unused (acellular) media sample at 50 C. The strongest acetaldehyde absorption peak has a peak-to-peak voltage of 2.2 mV, and the second strongest has a peak-to-peak voltage of 1.8 mV. These same peak voltages are obtained for a repeat measurement of the media sample after 5 days of freezer storage. The acetaldehyde concentrations provided by the strongest and second strongest peaks are 163 ppb and 180 ppb, respectively, giving an average concentration of 171 ppb. This concentration is about 33% greater than the acetaldehyde concentration of 129 ppb that was measured at a temperature of 37 °C.

Flow perfusion bioreactor and cancer cell cultures

The flow perfusion bioreactor used for this work consisted of six cylindrical chambers, each holding a collagen hydrogel, which provides a three-dimensional scaffold for cell attachment [21–25]. Cell culture media, drawn from a first reservoir by a peristaltic pump, flowed through gas permeable platinum-cured silicone tubing and the six chambers before returning to a second reservoir, which was connected to the first reservoir for replenishment. Approximately 70 ml of cell culture media contained in the two reservoirs circulated at a flow rate of 150 µL/min, which was low enough to prevent washout of the collagen hydrogel. A 3D printed mesh, fabricated from poly-L-lactic acid (PLLA) filament, was used to support the hydrogel. This mesh allowed media to flow around and into the hydrogel scaffold. Figure 7 shows a

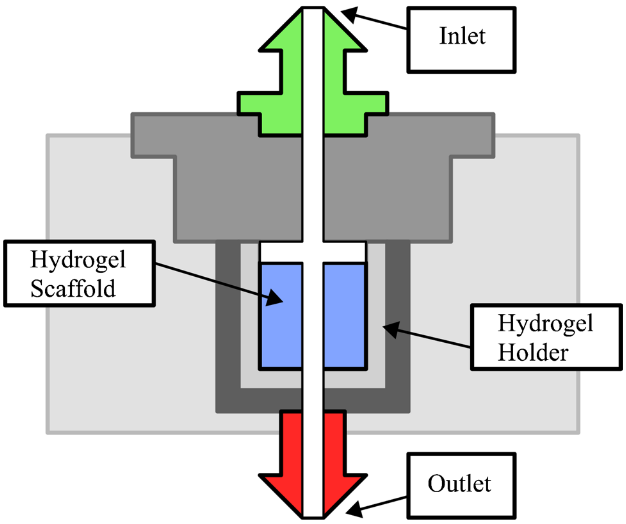


Fig. 7 Cross-sectional diagram of a single chamber in the bioreactor

cross-sectional drawing of a bioreactor chamber, and Fig. 8 shows photographs of an assembled bioreactor chamber and a 3D-printed mesh designed to hold a 7-mm diameter hydrogel scaffold.

EMT6 murine mammary carcinoma breast cancer cells were chosen for this work. EMT-6 cells are epithelial cells isolated from the mammary tumor of a mouse. They are widely used in cancer research as they form syngeneic triple negative tumors in mice, and they are known to express HIF-1α [26]. Prior to incorporation into the collagen hydrogel, the cells were incubated at 37 C and 5% CO_2 in a T75 culture flask with Waymouth's medium supplemented with 15% fetal bovine serum (FBS) and 1% antibiotic/antimycotic. After reaching 80% confluency, the cells were lifted with 0.05% trypsin–EDTA and centrifuged at 1100 relative centrifugal force (RCF) for 5 min to form a cell pellet. The supernatant was then discarded, and the cells were resuspended to form a cell solution. The collagen hydrogel consisted of rat tail collagen type I formed by mixing 9 parts of collagen solution with 1 part neutralizer solution according to the manufacturer's instructions. This 4 mg/mL collagen concentration was found to be the minimum needed for the hydrogel to resist damage from cell culture media flow. Breast cancer cell incorporation was performed by mixing 1 part cell solution with 4 parts neutralized collagen solution and repeatedly pipetting up and down. For each bioreactor run, six scaffolds were formed by pipetting 0.5 mL of the homogenized cell and collagen solutions into 6 wells of a 96-well plate and incubating for 1 h at 37 C and 5% CO_2. The bioreactor components were sterilized by autoclaving, except for the polymer components, which were sterilized for 12 h in an ethylene oxide system (Anprolene 74i, H.W. Andersen Products Ltd.) and allowed to outgas for at least 3 days in a sterile fume hood. The assembled bioreactor and cell culture media supplemented with 15% fetal bovine serum (FBS) and 1% antibiotic/antimycotic were then placed in the incubator set at 37 C and 5% CO_2 for at least 4 h. Tweezers were used to transfer the six hydrogel scaffolds, each containing breast cancer cells, to the bioreactor following its pre-conditioning in the incubator. Each cell culture run was performed for 4 days with continuously flowing media at 37 C and 5% CO_2.

Figure 9 shows photographs of a collagen hydrogel with incorporated breast cancer cells before transfer to a bioreactor chamber and the same collagen hydrogel scaffold after 4 days of culturing. Significant diameter reduction is observed, from 7 mm, obtained from the 96-well plate mold, to about 4 mm. Following each bioreactor run, the hydrogel scaffolds were placed in 3 mL of unsupplemented Waymouth's medium with 0.25% (w/v) collagenase type I for about 2 h, which was sufficient time for the collagenase to digest the collagen and free the breast cancer cells. This cell solution was transferred to a 15-mL falcon tube and an additional 2 mL of unsupplemented Waymouth's medium from container rinsing was added. A 100 μL aliquot of this cell solution was mixed with Trypan Blue (0.4% in solution) to stain the cells, and a 10 μL aliquot of this mixture was pipetted into a hemocytometer for microscopic cell counting. This same cell counting procedure was also used before the

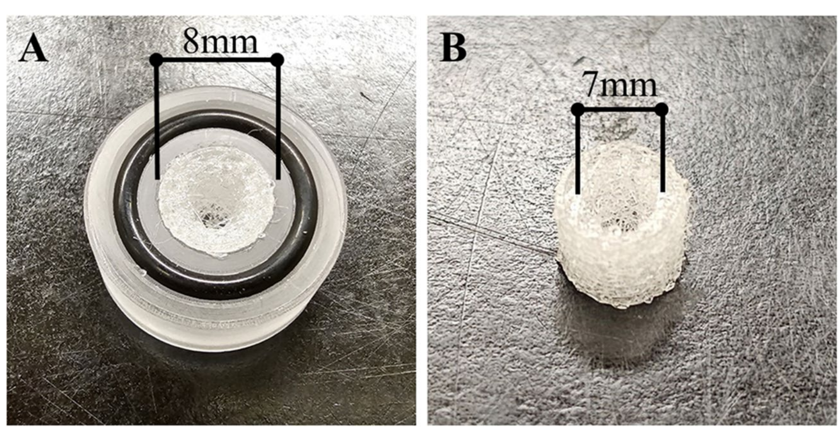


Fig. 8 The 3D-printed hydrogel holding mesh and rat tail type I collagen hydrogels. **A** Top-down view of the assembled bioreactor chamber, fabricated from Plexiglass, showing the o-ring and 3D-printed mesh. **B** 3D-printed PLLA mesh

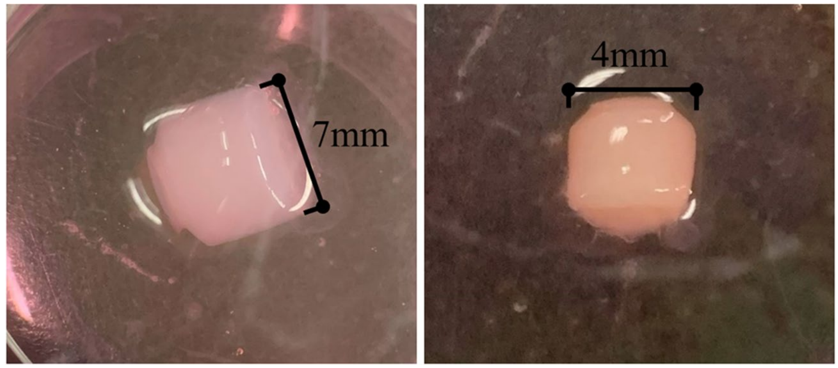


Fig. 9 Hydrogel scaffold before a 4-day run (left) and hydrogel scaffold after a 4-day culture in the bioreactor (right)

culture to obtain the quantity of breast cancer cells seeded in the collagen hydrogel. The cell culture media from each bioreactor run was transferred to a 50-mL falcon tube and frozen for storage prior to mid-IR laser measurement of acetaldehyde.

Normal bioreactor operation involves the use of oxygen-permeable tubing, with the cell culture media flowing in parallel through the 6 bioreactor chambers. The media entering each chamber is thus fully saturated with oxygen in this configuration. An alternative configuration involves the use of nonoxygen-permeable tubing, with the cell culture media flowing in series through the six bioreactor chambers. This daisy chain configuration (see Fig. 10), where the media do not reoxygenate until returning to the collection reservoir, results in oxygen-depleted media following oxygen consumption by the cells in upstream chambers. Oxygen consumption by cell metabolism will create increasingly hypoxic media, with the most hypoxic media flowing through the last chamber. A portion of the cells in the daisy chain (or series) configuration will therefore be cultured with hypoxic media. Bioreactor runs were performed with media flowing in both parallel and series configurations. An additional variant in cell culturing involved adding 150 μL of cobalt chloride ($CoCl_2$) to the media. $CoCl_2$ is known to stabilize HIF-1α activity when oxygen concentrations are at normal levels [27]. It simulates the effects of hypoxia by binding either to the iron center of HIF-hydroxylase, which degrades HIF-1α, or to HIF-1α directly, thus preventing its degradation [28].

Chemicals and materials used

Specific chemicals and materials used to perform the cell cultures were as follows: the gas permeable platinum-cured silicone tubing was the Masterflex brand from Cole Parmer, the EMT6 murine mammary carcinoma breast cancer cells were from ATCC, Manassas, VA, the cell culture media were Waymouth's medium from Gibco, the 15% fetal bovine serum (FBS)) supplement with 1% antibiotic/antimycotic was from Life Technologies, and the 0.05% trypsin–EDTA, the Trypan Blue (0.4% in solution), and the cobalt chloride ($CoCl_2$) were from Sigma-Aldrich.

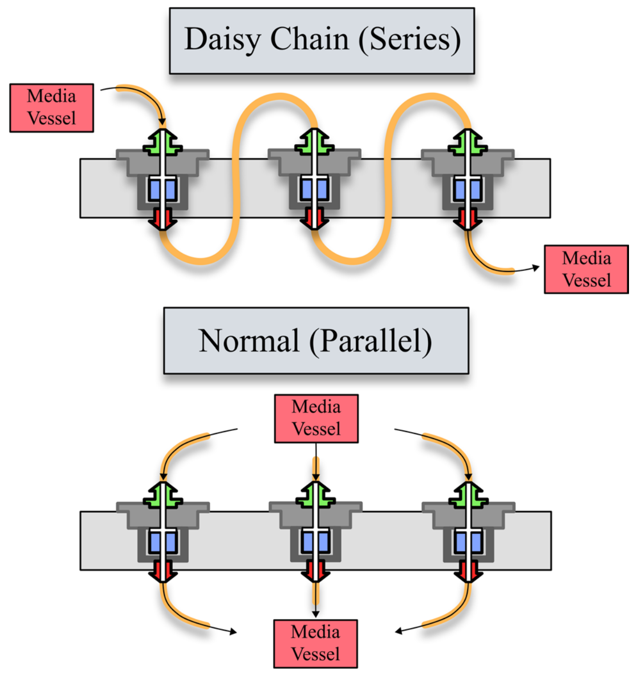


Fig. 10 Two configurations for media flow through the bioreactor chambers: (bottom) normal with oxygen-impermeable tubing and parallel flow and (top) daisy chain with nonoxygen-impermeable tubing and series flow

Experimental results

Cell culture measurements

Table 2 provides a list of headspace acetaldehyde concentration measurements for 11 different cell culture media samples, including the unused acellular media sample described above. All of the concentrations are for a 50 °C media sample temperature. The last column in Table 2 lists the percentage difference between the calculated acetaldehyde

Table 2 Summary of acetaldehyde concentration measurements for 11 different media samples. Sample 1 is an acellular media sample, and samples 2–11 are from 10 different cell culture runs. Cell counts before and after bioreactor culturing and percentage increase for each culture are listed. All headspace gas measurements were performed with the media sample at 50 C

Media sample	Seeded cell count (millions)	End cell count (millions)	Cell count increase (%)	Acetaldehyde					
				Peak 1	Peak 2	Peak 1	Peak 2	Average	Difference
				mV		ppb			%
1	--	--	--	2.2	1.8	164	180	172	9.4
2	3	6	100	5.7	4.0	425	400	412	6.0
3	6	10	67	6.8	5.2	507	520	513	2.6
380	6.6	11.3	71	5.1	4.0	380	400	390	5.1
5	27.6	30	8.7	2.3	1.6	171	160	166	6.9
6	10	18	80	11.0	8.1	820	810	815	1.2
7	10	8	− 20	--	--	--	--	--	--
8	10	14	40	2.5	1.8	186	180	183	3.4
9	10	13	30	1.3	1.2	97	120	108	21.4
10	5	9.3	86	4.4	4.2	328	420	374	24.7
11	10	11.6	16	7.0	5.5	522	550	536	5.3

concentrations for the two different absorption peaks. This value can be considered an indication of the quality of the acetaldehyde concentration measurement. A small difference indicates an absorption spectrum with absorption features characteristic of acetaldehyde, whereas a large difference indicates questionable acetaldehyde absorption features and a possibly unreliable acetaldehyde concentration measurement. Concentration disagreement can be due to system noise when acetaldehyde absorption is weak, optical absorption by other molecules in either spectral window for peak 1 or peak 2, and/or excessive optical interference fringes. All the acetaldehyde concentration measurements for peak 1 and peak 2 listed in Table 2 have concentration differences of less than 25% except for media sample 7, which will be discussed below.

Figure 11 shows measured spectra for media samples 2, 3, and 4. These three media samples were from successful cell cultures where cell counts increased by 100%, 67%, and 71%, respectively. Four spectra for each sample are shown, two at 40 C and two at 50 C as dashed and solid lines, saved about 1 min apart from each other. They all exhibit three clear absorption features for acetaldehyde. The peak voltages for the two strongest peaks, which are indicated for both sample temperatures (blue for 40 °C and red for 50 °C), give 50 C acetaldehyde concentrations of 412 ppb, 513 ppb, and 390 ppb, respectively.

Spectra for media sample 5 are shown in Fig. 12. Again, and in subsequent figures, two spectra in dashed and solid lines, saved about 1 min apart from each other, are shown for measurements with 40 C and 50 °C media sample temperatures. The 2.3 mV and 1.6 mV peak voltages give an average acetaldehyde concentration of 166 ppb. An unusual aspect of the cell culture performed with this media sample

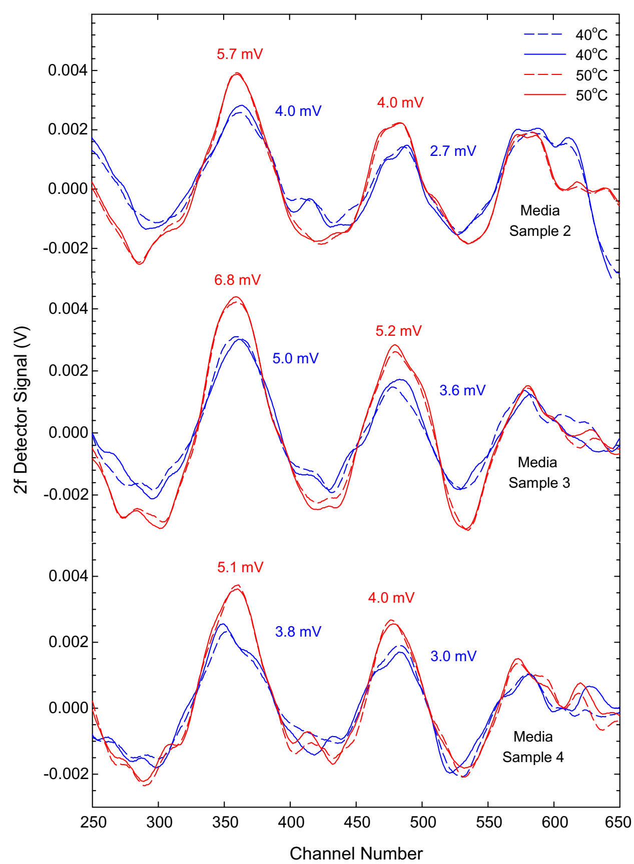


Fig. 11 Mid-IR laser absorption spectra for three media samples at 40 C and 50 C following four days of culturing cancer cells with cell count increases of 67% or more

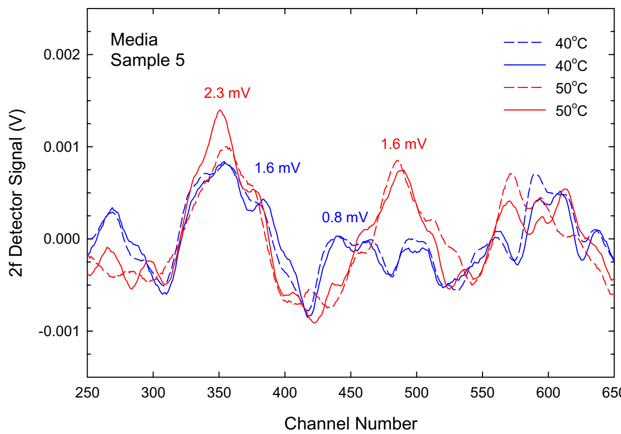


Fig. 12 Mid-IR laser absorption spectra for a media sample (number 5) at 40 C and 50 °C from a cell culture with a small percentage increase in cell count

was a high seeded cell count, 27.6 million cells, compared to the prior runs, which were 3.0, 3.6, and 6.6 million cells. As Table 2 shows, the percentage increase in cell count for media sample 5 after 4 days of culturing was the lowest, 8.7%. The lack of good cell replication with run number 5 indicates non-optimized cell culturing conditions, perhaps caused by cell overcrowding in the hydrogel scaffold.

Two sets of measurements were performed for media sample 6, which was obtained from a cell culture where the bioreactor was configured for the media to flow in series through each chamber. Figure 13 shows the peak-to-peak voltages for the strongest acetaldehyde absorption (peak 1), the second strongest (peak 2), and water absorption as a function of time during the headspace measurements of the media sample at 40 C and then at 50°. The first 10 min of the trend is for only chilled laboratory air flowing through the gas cell, after which lab air is bubbled through the 40 C media sample for about 4 min. After about 1 min, both acetaldehyde peaks begin increasing, reaching steady-state voltages of 10.7 mV and 7.9 mV for peaks 1 and 2, respectively. A second measurement at 50 C, showed peak-to-peak voltages of 11.6 mV and 10.3 mV for peaks 1 and 2, respectively. Acetaldehyde concentrations, using the peak-to-peak voltages obtained from the measured spectra were 641 ppb and 815 ppb at 40 C and 50 C, respectively. These were the highest headspace acetaldehyde concentrations measured among the cell culture media samples that were studied. The water peak in Fig. 13 shows the effect of the ice water cold trap. The peak-to-peak voltage for the water absorption feature near channel number 900 (see the lab air spectrum in Fig. 4) was at a steady 80 mV until the section of Teflon tubing immersed in ice water was removed to be blown dry. The peak-to-peak voltage for water increased to 197.8 mV a few minutes following removal and re-insertion of the cold trap. The water peak gradually decreased taking about 5 min

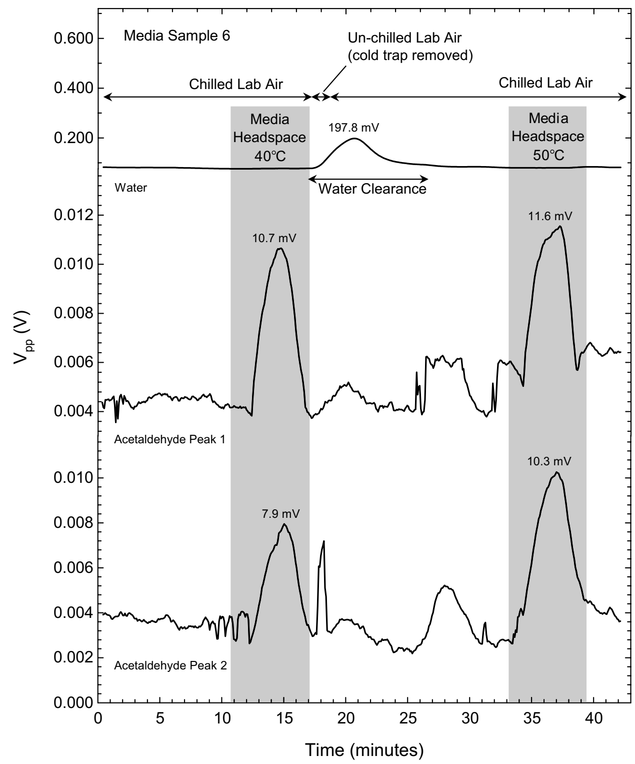


Fig. 13 Peak-to-peak voltage trends for water and the two strongest acetaldehyde peaks during 40 °C and 50 °C media sample 6 headspace measurements

to return to its steady state 80 mV level. At about this same time, both acetaldehyde peaks were observed to increase corresponding to an additional 200 ppb of acetaldehyde over a 2-min period of 100 ml/min air flow. Both acetaldehyde peaks fell to lower steady state values prior to the second headspace measurement at 50 C.

Media sample 6 was measured a second time following 9 days of freezer storage. Spectra for 40 C and 50 C media sample temperatures are shown in Fig. 14. The 50 C acetaldehyde concentration was 396 ppb, less than half as large as the concentration seen during the first measurement. A third measurement was performed after freezer storage for 11 months, and it showed 499 ppb; however, a fourth measurement performed the following day showed no measurable acetaldehyde. Peak-to-peak voltage trends for water and acetaldehyde (peak 2) are shown in Fig. 15. As with the measurements 9 days earlier (Fig. 13), the water peak increases when un-chilled lab air flows through the gas cell, which occurs twice following the two headspace measurements. The trend for acetaldehyde peak 2 shows increases in acetaldehyde concentration for the two headspace measurements along with an increase coinciding with the clearance of water from the gas cell. Repeat observation of this measurement artifact suggests a phenomenon related to an acetaldehyde/water interaction that traps acetaldehyde.

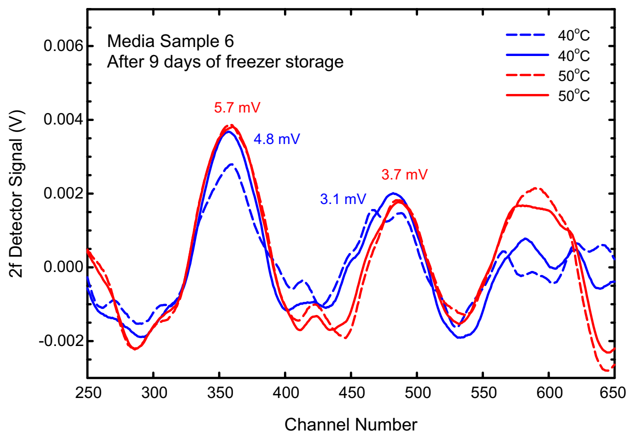


Fig. 14 Mid-IR laser absorption spectra from repeat measurements of a media sample (number 6) at 40 C and 50 C from a cell culture with in-series media flow to induce hypoxia in downstream chambers

These data show that about 10 min is needed between acetaldehyde measurements to prevent this effect from interfering with subsequent acetaldehyde concentration measurements. This was the only carry-over artifact observed during the measurements.

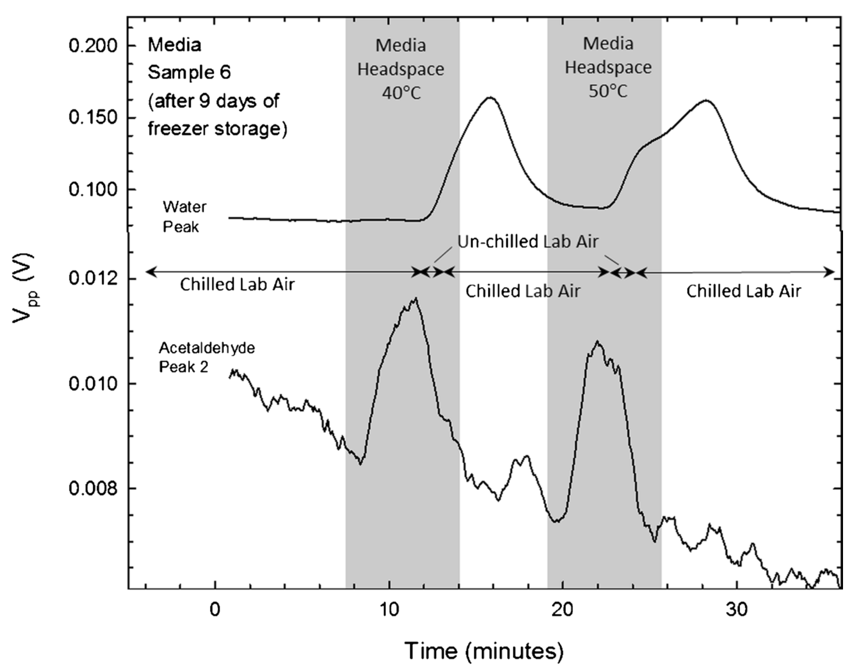
As Table 2 shows, media samples 7, 8, and 9 had either unmeasurable or very low acetaldehyde concentrations. Media sample 7 was from a cell culture in which the cell count at the end of the 4-day culture was 8 million cells, 2 million less than the seeded count of 10 million cells. This was the only bioreactor run where a decrease in cell count was observed. Measured spectra did not show the characteristic acetaldehyde absorption features, so reliable concentrations could not be obtained. Spectra for media sample 8 did show characteristic acetaldehyde absorption features, giving an acetaldehyde concentration of 183 ppb for a 50 C media sample temperature. The cell count for this culture only had a 40% increase, and it had an unusually cloudy appearance, suggesting that the culture was contaminated with foreign cells. Spectra for media sample 9 at 50 C yielded an acetaldehyde concentration of 108 ppb. This was the lowest acetaldehyde concentration measured for a media sample from a breast cancer cell culture.

Media sample 10, like media sample 6, was obtained from a cell culture performed with the bioreactor configured for the media to flow through the chambers in a daisy chain configuration. The acetaldehyde concentration for this sample was 374 ppb for a 50 C media sample temperature. The cell culture for media sample 11 was performed with cobalt chloride added to the media. Figure 16 shows spectra for this media sample at 40 C and at 50 C. The average acetaldehyde concentration from the two strongest acetaldehyde absorption peaks for the 50 C media sample is 536 ppb, with a disagreement of only 0.2%. This acetaldehyde concentration was the second highest observed among the 10 different breast cancer cell culture media samples that were measured, even though there was only a 16% increase in cell count.

Discussion

Overall, the experimental results presented in this paper illustrate the ability of mid-IR laser absorption spectroscopy to observe a characteristic set of rotational-vibrational

Fig. 15 Peak-to-peak voltage trends for water and acetaldehyde peak 2 during a second set of measurements of media sample 6 at 40 °C and 50°



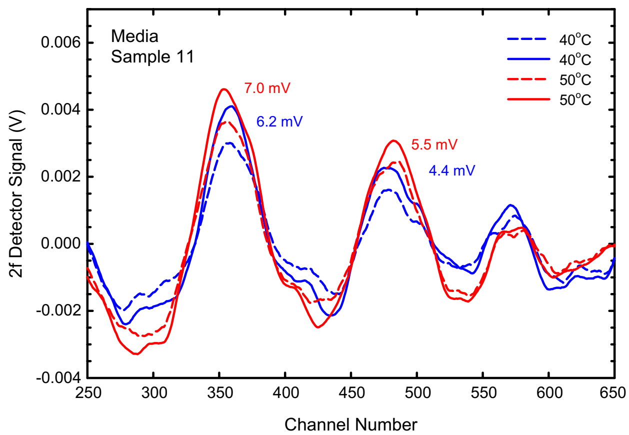


Fig. 16 Mid-IR laser absorption spectra for a media sample (number 11) at 40 C and 50 °C from a cell culture performed with cobalt chloride stabilization of HIF-1α

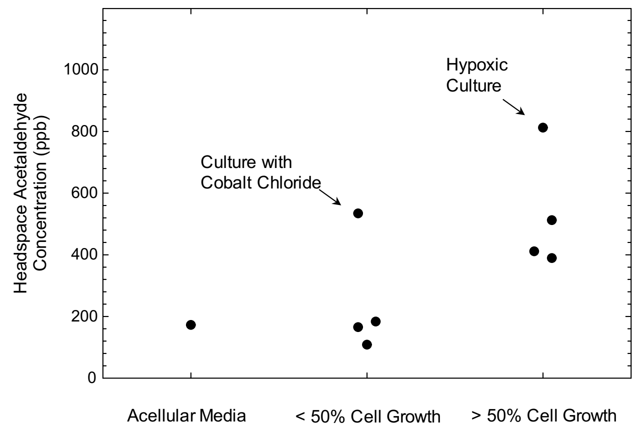


Fig. 17 Headspace acetaldehyde concentrations for three groups of cell culture media samples at 50 C

absorption peaks for acetaldehyde between 1770.20 cm^{-1} and 1770.35 cm^{-1}. These absorption peaks were strong for headspace gas samples from cancer cell culture media samples obtained from cultures with greater than 50% cell growth (see Figs. 11, 14, and 16), while cultures with less than 50% cell growth had weaker acetaldehyde absorption features, as can be seen in Fig. 12. Optical interference fringe noise due to standing waves in the optical cavities formed by reflective optics in the mid-IR laser spectrometer is the dominant noise in the spectra. The absorption spectra for acellular media headspace shown in Fig. 6 contain periodic peaks superimposed on weak acetaldehyde absorption peaks. With a spacing of 0.005 cm^{-1}, these Fabry–Perot resonant cavity peaks correspond to a roundtrip optical cavity length of 200 cm, a length that includes the Herriott gas cell length and spacings between the mirrors, laser, and detector on the optical bench. Interference fringe peaks are typically strong during the first 3 h of spectrometer operation, greater than 2 mV, but eventually become weaker, to less than 0.2 mV. This behavior is attributed to the optical bench reaching thermal stability following the powering up of the electronics instrumentation mounted below the optical bench. All measurements were performed following at least a 3-h warmup period because of this effect.

Gas-phase acetaldehyde concentrations were determined for 9 out of the 10 breast cancer cell culture media samples. The only media sample with a headspace acetaldehyde concentration below detection limits was from the only culture that did not show an increase in cell count, number 7. A summary of successful measurements is provided in Fig. 17. Three groups of data are shown, acellular media (number 1), cultures with less than 50% cell growth (numbers 5, 8, 9, and 11), and cultures with greater than 50% cell growth (numbers 2, 3, 4, 6, and 10). The measured acetaldehyde concentration in the headspace of an acellular media sample at 37 C was 129 ppb. This concentration is in good agreement with GC/MS [12] and SIFT-MS [9, 10] measurements, which were between 110 and 260 ppb. The acetaldehyde concentrations for cultures with less than 50% cell growth were clustered around 150 ppb, except for number 11, the culture with HIF-1α stabilizing cobalt chloride, which had an acetaldehyde concentration of 536 ppb. For cultures with greater than 50% cell growth, the acetaldehyde concentrations were clustered around 440 ppb, except for number 6, a culture with hypoxic media, which had an acetaldehyde concentration of 815 ppb. These results show that mid-IR laser spectroscopy can reliably measure acetaldehyde concentrations in the headspace gas of cell culture media samples and that these concentrations can be at high levels when there is hypoxia or cobalt chloride stabilization of HIF-1α activity.

Figure 18 plots headspace acetaldehyde concentration as a function of media sample temperature. Measurements of seven different media samples from breast cancer cell cultures at 40 C and 50 C are shown along with the vapor pressure versus temperature data for acetaldehyde. Extrapolation of the 40 C and 50 C data along the vapor pressure trend gives acetaldehyde concentrations between 250 and 600 ppb for a media sample temperature of 37 C. These acetaldehyde concentrations can be compared to GC/MS and SIFT-MS measurements of leukemia and lung cancer cell cultures where headspace gas sampling was performed at 37 C. As can be seen in Table 3, the range of acetaldehyde concentrations are similar: 611 ppb to 785 ppb for leukemia cells and 82 ppb to 1973 ppb for lung cancer cells. In general, high headspace acetaldehyde concentrations could be obtained with much smaller seeded quantities of cells when hydrogel scaffolds were used. For example, media sample 6, which was seeded with 10 million breast cancer cells, had an extrapolated 37 C headspace acetaldehyde concentration of 590 ppb, and ref. [11] describes a hydrogel scaffold seeded with only 1 million lung cancer cells, producing a headspace

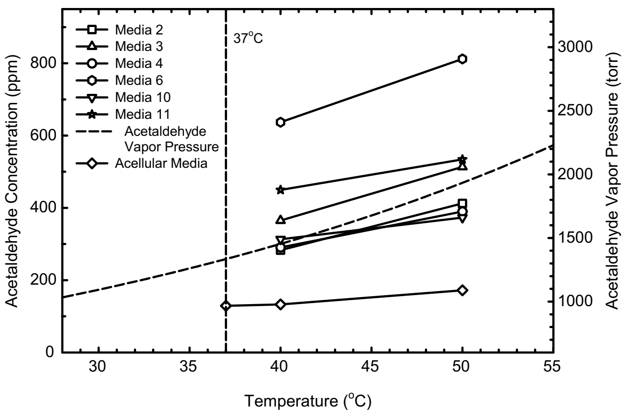


Fig. 18 Acetaldehyde concentrations in the headspace of six different media samples as a function of media sample temperature, along with acetaldehyde vapor pressure as a function of temperature

acetaldehyde concentration of 1973 ppb following 16 h of culturing in a sealed flask. By contrast, suspension cultures needed more than 100 million seeded cells to produce more than 1000 ppb of acetaldehyde. These results are consistent with more effective nutrient consumption and gas exchange when 3D scaffolds are used.

The data summarized in Table 3 also show that culturing breast cancer cells in a flow perfusion bioreactor resulted in an increase in cell count, whereas the culturing of leukemia and lung cancer cells in a sealed flask without flowing media resulted in decreased cell counts, where survival rates were between 40 and 95%. By contrast, 9 of the 10 breast cancer cell cultures in a flow perfusion bioreactor had an increase in cell count from a low of 8.7% to a high of 100%, and this increase correlates with acetaldehyde concentration, as shown in Fig. 17. This comparison shows that a flow perfusion bioreactor, with cells cultured in hydrogel scaffolds, provides a potentially more accommodating environment for cellular proliferation and thus acetaldehyde production. Due to cell overcrowding, culture number 5 likely experienced a lack of good nutrient supply, and it had the lowest increase in cell count and a relatively low acetaldehyde concentration.

With cell culture media contained in reservoirs separated from the cultured cells, the configuration of a flow perfusion bioreactor allows headspace gas measurements without disturbing the cell culture. Headspace gas samples during culturing can be obtained by flowing a carrier gas, either filtered lab air or cylinder gas(es), through a media reservoir and passing the exhaust through the gas cell of a mid-IR laser spectrometer. The acetaldehyde concentration trends in Figs. 13 and 15 show the type of data that can be produced with such headspace gas sampling. Measurements can be done periodically during a culture, lasting about 5 min each time, or they can be performed continuously. Future work in this area should focus on integrating a mid-IR laser spectrometer with a flow perfusion bioreactor for on-line, real-time measurements of headspace acetaldehyde concentrations. This will offer an attractive platform for studying cancer cell metabolism.

Other molecules besides acetaldehyde have absorption features within the 1770.20 cm^{-1} to 1770.35 cm^{-1} spectral region covered by the mid-IR laser used in this work. They include ammonia, acetic acid, formic acid, acetone, ethanol, and methanol. Except for ammonia, no evidence for the presence of these other molecules in the headspace of media samples from breast cancer cell cultures was found. This observation does not necessarily exclude these other molecules at concentrations that are below the detection limits, which are likely relatively high since this spectral region does not coincide with strong absorption bands for these other molecules. Ammonia, which has a rotational-vibrational absorption line at 1770.31 cm^{-1}, was observed when the media sample temperatures exceeded 60 C. The source of ammonia in this case is most likely from thermal deamination of amino acids in the cell culture media. GC/MS measurement of media headspace samples from cultured human leukemia cells showed that the concentrations for acetaldehyde were much larger than those for other measured headspace gases. For example, the next most abundant molecule observed to increase with HL60 leukemia cells was hexanaldehyde. Its concentration was 2 ppb in the headspace of acellular media and 11 ppb in the headspace of

Table 3 Summary of acetaldehyde concentration measurements in the headspace gas of different cancer cell cultures. Mid-IR laser acetaldehyde concentration values are extrapolated to 37 C from 40 C and 50 C measurements

Technique	Ref	Cancer cells	Seeded quantity (million)	Culture technique and time	Cell count increase (decrease) (%)	Headspace acetaldehyde concentration (ppb)
Mid-IR Laser	This work	Breast (EMT-6)	3.0–27.6	3D scaffold with flow perfusion 4 days	9–100	250–600
GC/MS	[12]	Leukemia (HL60)	40	Suspended 48 h	(24–36)	611–785
SIFT-MS	[9]	Lung (CALU-1)	40–120	Suspended 16 h	(20–60)	273–1470
		Lung (SK-MES)	50–200	Suspended 16 h	(10–20)	82–854
	[10]	Lung (CALU-1)	50–80	Suspended 16 h	(5–7)	387–545
	[11]	Lung (CALU-1)	0.01–40	3D scaffold 16 h	-	176–1973

cell culture media after 48 h of culturing [12]. SIFT-MS measurements also showed that ethanol and acetic acid are present in the headspace of media samples used for culturing lung cancer cells, but at concentrations significantly lower than those measured for acetaldehyde [9–11].

It is clear from the mid-IR laser measurement results presented here, along with prior GC/MS and SIFT-MS measurements, that acetaldehyde is produced by cancer cells and that its headspace concentration can provide information about cancer cell metabolism. As discussed above, acetaldehyde is a product of pyruvate decomposition, and its thermodynamic properties make it a favorable molecule for detection in the gas phase. In aqueous solutions, it has a high activity coefficient, in the range of 80 [8], so once it is produced in cell cytosol, it will rapidly enter the gas phase. A change in cell metabolism that affects cytosol pyruvate concentration should therefore be rapidly reflected as a change in the acetaldehyde concentration in headspace gas. This effect makes headspace acetaldehyde concentration measurements useful for monitoring cell metabolism. For example, adding glucose to cell culture media will increase pyruvate concentration in cell cytosol, and this increase can be monitored by observing an increase in headspace acetaldehyde concentration. In addition, the blockage of pyruvate entry into the mitochondria for aerobic respiration, a result of HIF-1α activity, will also result in an increase in cytosol pyruvate concentration and a corresponding increase in headspace acetaldehyde concentration. The results presented here show this effect. Higher acetaldehyde concentrations were observed for cell cultures performed with cell culture media flowing serially through the six bioreactor chambers, in which case the cell cultures in the downstream chambers experienced lower oxygen concentrations, thus increasing HIF-1α activity (media sample 6). Higher acetaldehyde concentrations were also observed for cell cultures performed with cobalt chloride, which stabilizes HIF-1α under normoxic conditions (media sample 11).

Continuous monitoring of acetaldehyde concentration during media sample headspace gas sampling, see Figs. 13 and 15, showed the occurrence of a measurement artifact where an emission of acetaldehyde was observed at the same time that water concentration decreased to a steady-state level following condensate clearance from a cold trap. This phenomenon, which was observed twice, is not currently completely understood but appears to be related to acetaldehyde interaction with water on surfaces. Obtaining reliable acetaldehyde concentration measurements will require the elimination of such artifacts. Measurement methods should therefore be used that allow continuous monitoring with steady-state air flows and stable background water concentrations. The effects of media sample storage and repeat measurements were also studied. For example, the first two measurements of media sample 6 were performed 9 days apart with freezer storage between the measurements. A significant decrease in acetaldehyde was observed between the first and second measurements, from 815 to 396 ppb. A third 50 C measurement was performed after freezer storage for 11 months, and it showed a 499 ppb acetaldehyde concentration, while a fourth measurement performed the following day showed no measurable acetaldehyde. These results suggest that freezer storage is effective in stabilizing dissolved acetaldehyde; however, they also show that there can be a significant acetaldehyde loss during each measurement. Reducing media measurement temperature from 50 to 37 C, a requirement for on-line measurements, will reduce this effect. Nevertheless, these results show that acetaldehyde can be lost due to handling procedures, thus highlighting the need for consistent sample handling procedures when performing off-line measurements.

With this work, mid-IR laser absorption measurement of acetaldehyde has now been demonstrated in two spectral regions of the carbonyl stretch band as shown in Fig. 19. Our prior work [13] was performed with a IV-VI semiconductor diode laser on the low frequency side of the P branch, near 1727 cm^{-1}, and the work described here was performed with an ICL on the high frequency side of the R branch, near 1770.25 cm^{-1}. This spectrometer had an acetaldehyde minimum detection limit of 37 ppb, very similar to the 40 ppb acetaldehyde detection limit for the prior spectrometer. This result is significant because the earlier work was performed with both the mid-IR laser and detector cooled to cryogenic temperatures, $T < 100$ K, while this current work was performed with both the ICL and detector cooled with thermoelectric modules. This demonstrates that mid-IR laser spectrometers can be compact and easy to use. As shown here, they can provide real-time, reagent-free acetaldehyde concentration measurements in gas-phase samples. Further development of ICL sensor technology can focus on integrating acetaldehyde sensors with bioreactors for on-line monitoring of cell culture headspace. Work can also focus on specific cancer detection research projects such as exhaled breath measurements for lung cancer detection. In this case, the minimum detection limit for acetaldehyde will need to be improved by about a factor of two so that the median 24 ppb acetaldehyde concentrations in the exhaled breath of healthy humans [29] can be detected. Such improvement is possible by using stronger rotational-vibrational absorption lines than those used here. Based on the FTIR absorption spectrum for acetaldehyde (see Fig. 19), absorption lines twice as strong are likely to be found near the peaks of the P, Q, and R branches of the carbonyl stretch band, between 1730 and 1760 cm^{-1}. ICLs with emission in these spectral regions can be used to build compact sensors for reliable acetaldehyde concentration measurements in

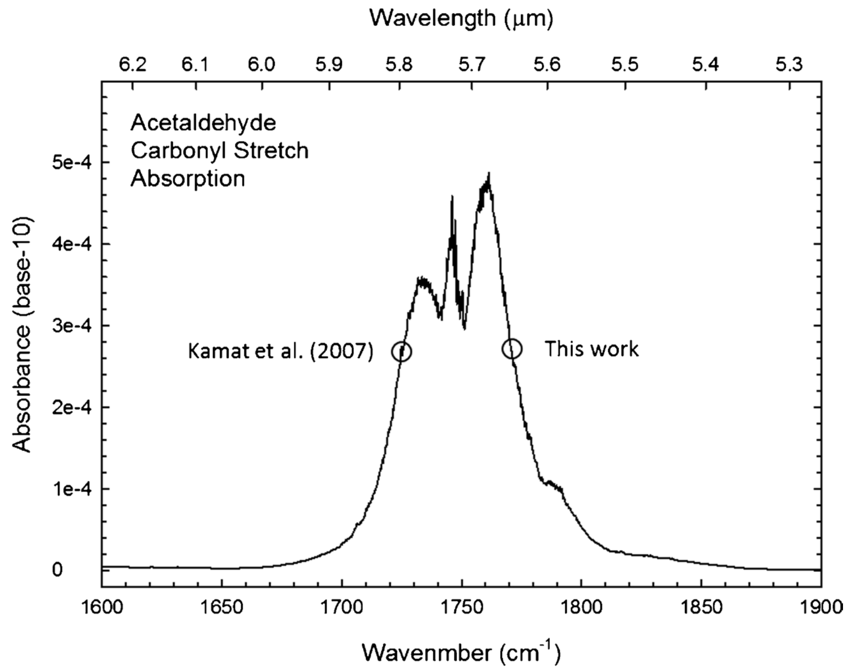


Fig. 19 FTIR absorption spectrum for the carbonyl stretch band of acetaldehyde. Two spectral regions where high spectral resolution mid-IR laser absorption spectroscopic measurements have been performed [13] are indicated

human breath. Such sensors will facilitate future projects focused on measuring acetaldehyde in exhaled breath for cancer detection research.

Additional future work on improving mid-IR laser spectrometer performance can focus on deconvolving optical interference fringe noise from analyte absorption features. Machine learning algorithms can be used, for example, to recognize the periodic nature of interference fringes, as well as to recognize the broader absorption features of rotational-vibrational molecular modes of motion. Mid-IR laser spectroscopy is well suited for this machine learning application. Data are provided at a high rate, 1000 channels of data every 0.1 s, allowing efficient generation of labeled datasets for neural network training. For example, large datasets of absorption spectra for gas samples with different analyte calibrations and different gas mixture compositions, as well as datasets for noise cancellation applications, can be generated in a short amount of time. Development of such machine learning techniques will improve spectrometer performance. For example, as seen in Fig. 13, the abrupt jumps in the peak-to-peak voltage trends between 7 and 20 min for acetaldehyde peak 2 and between 25 and 35 min for acetaldehyde peak 1 are likely due to optical interference fringes drifting through the acetaldehyde absorption peak windows. Machine learning techniques can remove these interferences and help make acetaldehyde concentration measurements more accurate. The work described in this paper is important because it shows that a mid-IR laser technology can selectively measure acetaldehyde in cancer cell culture media headspace, thus providing a new way to observe cell metabolism, with near-term applications in improving bioreactor process control.

Summary

This paper has documented a new way to observe acetaldehyde, a gas-phase byproduct of cancer cell metabolism. Experimental work involved the development and use of a mid-IR laser absorption spectrometer designed to measure acetaldehyde using mid-IR laser absorption within the carbonyl stretch band, near a photon wavelength of 5.65 µm, and using this spectrometer to measure cell culture media samples obtained from breast cancer cell cultures in a flow perfusion bioreactor. Off-line measurements were performed by bubbling lab air through about 40 ml of cell culture media and flowing the exhaust gas through a long optical path gas cell, where rotational-vibrational modes of acetaldehyde were excited by the laser. Acetaldehyde concentrations were determined from peak-to-peak voltages for the two strongest acetaldehyde absorption features within a spectral region spanning 1770.20 to 1770.35 cm^{-1}. Three different media samples obtained from breast cancer cell cultures with the same nominal culturing conditions, all measured at 50 C, had headspace acetaldehyde concentrations of 412 ppb, 513 ppb, and 390 ppb, while two media samples with imposed hypoxia (media sample 6) or cobalt chloride stabilization of HIF-1α activity (media sample 11) had higher acetaldehyde concentrations, 815 ppb and 536 ppb, respectively. These results establish experimental proof of concept for the use of mid-IR laser absorption spectroscopy to monitor cell culture metabolism through the measurement of acetaldehyde concentrations in cell culture headspace gas.

Multiple examples of laser absorption spectra, along with real-time acetaldehyde concentration trends, are provided in this paper. They document quantifiable acetaldehyde concentration measurements above a minimum detection limit of 37 ppb, a sensitivity that is sufficient to measure the background acetaldehyde concentration in acellular cell culture media headspace, as seen in Fig. 6, and easily observe elevated acetaldehyde concentrations in the headspace of cancer cell culture media samples. This acetaldehyde sensing capability was provided, for the first time, with a thermoelectrically cooled mid-IR laser, making this technology currently suitable for on-line headspace gas sampling of cell cultures and much simpler to operate than GC/MS or SIFT-MS instrumentation. Expanding acetaldehyde sensing applications to exhaled breath research will require an improved minimum detection limit such that baseline acetaldehyde concentrations in the breath of healthy humans, 24 ppb [29], can be measured. In addition to using stronger absorption lines to improve sensitivity, ICL sensor sensitivity can also be improved by use of longer optical path gas cells [30]. Future work focused on developing more sensitive ICL sensors will allow this compact and easy-to-use technology to be used for research on cancer detection using exhaled breath analysis, for example.

Author contribution Bryant: investigation, methodology, validation, writing (original draft). McCann: conceptualization, funding acquisition, investigation, data curation, writing (review and editing). Namjou: investigation, methodology, writing (original draft). Sikavitsas: funding acquisition, supervision, writing (review and editing). Harrison: conceptualization, resources, writing (review and editing).

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Data availability Data are available from the authors by request.

Declarations

Conflict of interest The authors declare no competing interests.

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